

ENGR 102 Summer Bridge

Day 11 Student Workbook

While Loops, Update Order, Sentinels, and Termination

Monday, July 20, 2026 • 50 points

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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Daily target

increasing independence

Design loops whose state changes make termination demonstrable.

Allowed tools	Prior tools plus while loops and sentinel-controlled input
Support supplied	The sentinel, valid-data interval, and required summary are supplied.
You must decide	Loop pattern, update placement, validity branch, and the empty-data case.
Scaffold level	Termination evidence prompted; algorithm no longer sequenced line by line.

Scoring and completion standard

50 points

Evidence	Points	Secure work shows
Integrated trace	12	intermediate state, condition results, and exact output
Diagnosis and repair	10	actual, intended, cause, repair, and a boundary test when requested
Full program and tests	28	coherent executable logic, required output, and defended tests

Independence standard. You may use the supplied requirements and examples, but you must produce one connected solution. A collection of disconnected correct lines is not yet a complete program.

Begin with the trace, then use its evidence habits when you construct.

Task 1. Integrated trace**12 points**

Trace both parts of the compound condition before each pass and the state after each update.

```
level = 17
steps = 0

while level < 50 and steps < 4:
    level = level + 9
    steps = steps + 1
    print(level, steps)

print("final", level, steps)
```

Your task

1. Build one ledger with condition result, level, and steps before and after each pass.
2. Give every output line and explain which condition evidence proves termination.

Task 2. Diagnose and repair**10 points**

This loop should read integers until -1 and count nonnegative entries, but it can repeat forever.

```
reading = int(input())
valid_count = 0

while reading != -1:
    if reading >= 0:
        valid_count = valid_count + 1

print(valid_count)
```

Your task

1. Explain the infinite-loop condition using the unchanged state, not just the phrase missing update.
2. Write the minimal repair and describe the final false condition for inputs 5, 8, -1.

Task 3. Construct a complete program**28 points across Pages 4-5****Program specification**

- Read integer readings until the sentinel -1 is entered.
- Values from 0 through 100 inclusive are valid. Any other non-sentinel value is ignored and must print Ignored.
- For valid values, compute count, total, and the number at or above 70.
- After termination, print count, total, and high_count on separate lines. Then print the average, or print No data when count is zero.

Required planning evidence

1. State the loop-controlling variable and show where it changes so that every path either terminates or reaches another input operation.

Program workspace – begin the connected solution here.

Task 3 continued. Test and defend the program**included in 28 points**

Continue code if needed.

Required tests, expected results, and brief defense

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VIP Lab Alignment

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This page adds a current lab handoff while preserving the workbook's independence-ramp tasks.

Why this page is here

VIP Lab Day(s): 5

Activities: 18.19, 18.21, 18.22

Complete the same while-loop cells with less planning support and defend the empty or already-finished case.

Open these current notebook cells

Activity / stable cells	Notebook work	Evidence to carry forward
18.19 18-19-work / 18-19-transfer / 18-19-cold	Countdown To Multiple	Write a while condition from a divisibility goal.
18.21 18-21-work / 18-21-transfer / 18-21-cold	Cooldown Sequence	Track a current value and a stored sequence.
18.22 18-22-work / 18-22-transfer / 18-22-cold	Rounds To Clear Queue	Model repeated queue reduction.

Readiness check before you code

1. Predict whether each loop executes zero, one, or multiple times for its transfer case before running it. _____

2. Identify where output or list append must occur relative to the update. _____

Notebook and submission procedure

1. Use the sample and transfer cells for formative practice; attempt before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only the Studio cells listed above are scored for independent mastery in the matching Lab Day notebook.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.